

[4.2.7]
[6.6.1]
[9.4.2]

Theorem 3.3.6. (Global Version of Menger's Theorem)

- (i) A graph is k -connected if and only if it contains k independent paths between any two vertices.
- (ii) A graph is k -edge-connected if and only if it contains k edge-disjoint paths between any two vertices.

Proof. (i) If a graph G contains k independent paths between any two vertices, then $|G| > k$ and G cannot be separated by fewer than k vertices; thus, G is k -connected.

Conversely, suppose that G is k -connected (and, in particular, has more than k vertices) but contains vertices a, b not linked by k independent paths. By Corollary 3.3.5 (i), a and b are adjacent; let $G' := G - ab$. Then G' contains at most $k - 2$ independent a - b paths. By Corollary 3.3.5 (i), we can separate a and b in G' by a set X of at most $k - 2$ vertices. As $|G| > k$, there is at least one further vertex $v \notin X \cup \{a, b\}$ in G . Now X separates v in G' from either a or b – say, from a . But then $X \cup \{b\}$ is a set of at most $k - 1$ vertices separating v from a in G , contradicting the k -connectedness of G .

(ii) follows straight from Corollary 3.3.5 (ii). $\square$

a, b
 G'
 X
 v

3.4 A -paths and Mader's theorem

In analogy to Menger's theorem we may consider the following questions. Given a set A of vertices in a graph, how many disjoint, edge-disjoint, or independent A -paths can we find in that graph? Is the largest number of such paths related to the least size of a vertex or edge cover of all its A -paths?

An early theorem of Gallai goes in this direction. In Erdős-Pósa spirit – see Chapter 2.3 – it says that if a graph fails to contain many disjoint A -paths it has a small set of vertices that covers all its A -paths, and thus 'totally disconnects' A :

Theorem 3.4.1. (Gallai 1961)

Let A be a set of vertices in a graph G , and $k \geq 1$ an integer. Then G either contains k disjoint A -paths or has a set of at most $2k - 2$ vertices that meets every A -path.

The graph K^{2k-1} , with A its entire vertex set, shows that $2k - 2$ is lowest possible as a general bound, one that depends only on k .

Theorem 3.4.1, as well as its analogue for edge-disjoint paths (with the same bound of $2k - 2$), will be an easy consequence of Mader's theorem below, via its Corollary 3.4.3. See Exercises 25 and 30.

Mader's theorem solves our initial problem for independent H -paths rather than A -paths. Recall that, for $H \subseteq G$, an H -path in G is an A -path for $A = V(H)$ that is not just a single edge of H . If H is induced in G , as it will be for Mader's theorem, the H -paths in G are thus precisely its A -paths of length at least 2, for $A = V(H)$.

In addition to implying Gallai's theorem and its edge-analogue, Mader's theorem also implies a similar statement about independent A -paths: just subdivide any A -paths of length 1, which turns them into H -paths (with $V(H) = A$) without changing the number of A -paths.

In the fashion of Menger's theorem, Mader's theorem observes an upper bound on the number of independent H -paths that arises naturally from the size of certain separators, and then states that this bound is always attained by some set of paths.

What could such an upper bound look like? Clearly, if $X \subseteq V(G - H)$ and $F \subseteq E(G - H)$ are such that every H -path in G has a vertex or an edge in $X \cup F$, then G cannot contain more than $|X \cup F|$ independent H -paths. Hence, the least cardinality of such a set $X \cup F$ is a natural upper bound for the maximum number of independent H -paths. (Note that every H -path meets $G - H$, because H is induced in G and edges of H do not count as H -paths.)

In contrast to Menger's theorem, this bound can still be improved. The minimality of $|X \cup F|$ implies that no edge in F has an end in X : otherwise this edge would not be needed in the separator. Let $Y := V(G - H) \setminus X$, and denote by $\mathcal{C}_F$ the set of components of the graph (Y, F) . Since every H -path avoiding X contains an edge from F , it has at least two vertices in ∂C for some $C \in \mathcal{C}_F$, where ∂C denotes the set of vertices in C that send an edge of G to $G - X - C$ (Fig. 3.4.1).

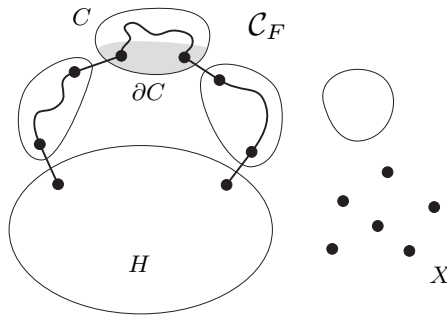


Fig. 3.4.1. An H -path in $G - X$

The number of independent H -paths in G is therefore bounded above by

$$M_H(G) := \min \left(|X| + \sum_{C \in \mathcal{C}_F} \left\lfloor \frac{1}{2} |\partial C| \right\rfloor \right), \quad M_H(G)$$

where the minimum is taken over all X and F as described above:

X $X \subseteq V(G - H)$ and $F \subseteq E(G - H - X)$ such that every H -path in G has a vertex or an edge in $X \cup F$.

Mader's theorem says that this upper bound is always attained by some set of independent H -paths:

Theorem 3.4.2. (Mader 1978)

Given a graph G with an induced subgraph H , there are always $M_H(G)$ independent H -paths in G .

In order to obtain more immediate analogues to the vertex and edge version of Menger's theorem, let us consider the two special cases of the above problem where either F or X is required to be empty. Given an induced subgraph $H \subseteq G$, we denote by $\kappa_H(G)$ the least cardinality of a vertex set $X \subseteq V(G - H)$ that meets every H -path in G . Similarly, we let $\lambda_H(G)$ denote the least cardinality of an edge set $F \subseteq E(G)$ that meets every H -path in G .

$\kappa_H(G)$

$\lambda_H(G)$

Corollary 3.4.3. Given a graph G with an induced subgraph H , there are at least $\frac{1}{2}\kappa_H(G)$ independent H -paths and at least $\frac{1}{2}\lambda_H(G)$ edge-disjoint H -paths in G .

k *Proof.* To prove the first assertion, let k be the maximum number of independent H -paths in G . By Theorem 3.4.2, there are sets $X \subseteq V(G - H)$ and $F \subseteq E(G - H - X)$ with

$$k = |X| + \sum_{C \in \mathcal{C}_F} \lfloor \frac{1}{2} |\partial C| \rfloor$$

such that every H -path in G has a vertex in X or an edge in F . For every $C \in \mathcal{C}_F$ with $\partial C \neq \emptyset$, pick a vertex $v \in \partial C$ and let $Y_C := \partial C \setminus \{v\}$; if $\partial C = \emptyset$, let $Y_C := \emptyset$. Then $\lfloor \frac{1}{2} |\partial C| \rfloor \geq \frac{1}{2} |Y_C|$ for all $C \in \mathcal{C}_F$. Moreover, for $Y := \bigcup_{C \in \mathcal{C}_F} Y_C$ every H -path has a vertex in $X \cup Y$. Hence

Y

$$k \geq |X| + \sum_{C \in \mathcal{C}_F} \frac{1}{2} |Y_C| \geq \frac{1}{2} |X \cup Y| \geq \frac{1}{2} \kappa_H(G)$$

as claimed.

The second assertion follows from the first by considering the line graph of G (Exercise 27). $\square$

It may come as a surprise to see that the bounds in Corollary 3.4.3 are best possible (as general bounds): one can find examples for G and H where G contains no more than $\frac{1}{2}\kappa_H(G)$ independent H -paths or no more than $\frac{1}{2}\lambda_H(G)$ edge-disjoint H -paths (Exercises 28 and 29).

3.5 Linking pairs of vertices

Let G be a graph, and let $X \subseteq V(G)$ be a set of vertices. We call X *linked* in G if whenever we pick distinct vertices $s_1, \dots, s_\ell, t_1, \dots, t_\ell$ in X we can find disjoint paths $P_1, \dots, P_\ell$ in G such that each P_i links s_i to t_i and has no inner vertex in X . Thus, unlike in Menger's theorem, we are not merely asking for disjoint paths between two sets of vertices: we insist that each of these paths shall link a specified pair of endvertices.

linked

If $|G| \geq 2k$ and every set X of at most $2k$ vertices is linked in G , then G is *k-linked*. Clearly, this is equivalent to requiring merely that $|G| \geq 2k$ and disjoint paths $P_i = s_i \dots t_i$ exist for every choice of exactly $2k$ distinct vertices $s_1, \dots, s_k, t_1, \dots, t_k$: just add dummy vertices to X to bring it up to size $2k$. In practice, the latter is easier to prove, because we need not worry about inner vertices in X .

k-linked

Clearly, every k -linked graph is k -connected. The converse, however, seems far from true: being k -linked is clearly a much stronger property than k -connectedness. Still, we shall prove in this section that we can force a graph to be k -linked by assuming that it is $f(k)$ -connected, for some function $f: \mathbb{N} \rightarrow \mathbb{N}$. We first borrow a lemma from Chapter 7 to give a nice and simple proof that such a function f exists at all. In the remainder of the section we then prove that f can even be chosen linear.

The basic idea in the simple proof is as follows. If we can prove that G contains a subdivision K of a large complete graph, we can use Menger's theorem to link the vertices of X disjointly to branch vertices of K , and then hope to pair them up as desired through the subdivided edges of K . This requires, of course, that our paths do not hit too many of the subdivided edges before reaching the branch vertices of K .

The lemma saying that large enough connectivity does indeed force the existence of such a complete topological minor K will be proved in Chapter 7.2, where we consider several results of this type. By Theorem 1.4.3 it suffices to assume that G has large average degree:

Lemma 3.5.1. *There is a function $h: \mathbb{N} \rightarrow \mathbb{N}$ such that, for every $r \in \mathbb{N}$, every graph of average degree at least $h(r)$ contains K^r as a topological minor.*

Theorem 3.5.2. (Jung 1970; Larman & Mani 1970)

There is a function $f: \mathbb{N} \rightarrow \mathbb{N}$ such that every $f(k)$ -connected graph is k -linked, for all $k \in \mathbb{N}$.

Proof. We prove the assertion for $f(k) = h(3k) + 2k$, where h is a function as in Lemma 3.5.1. Let G be an $f(k)$ -connected graph. Then $d(G) \geq \delta(G) \geq \kappa(G) \geq h(3k)$; let K be a TK^{3k} in G as provided by Lemma 3.5.1, and let U denote its set of branch vertices.

(3.3.1)

 G K U

For the proof that G is k -linked, let distinct vertices $s_1, \dots, s_k$ and $t_1, \dots, t_k$ be given. By definition of $f(k)$, we have $\kappa(G) \geq 2k$.

 s_i, t_i